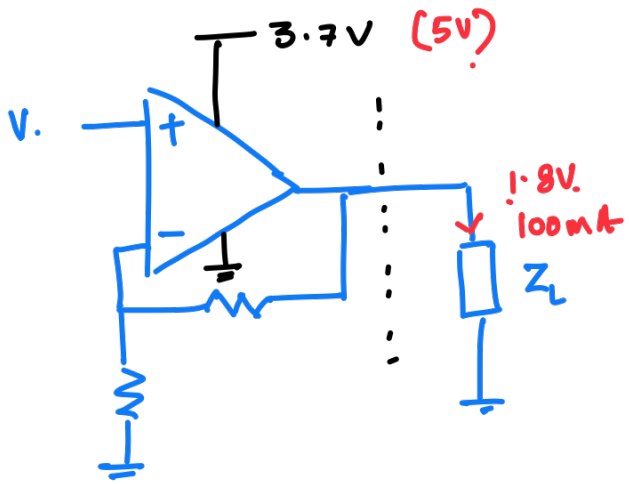
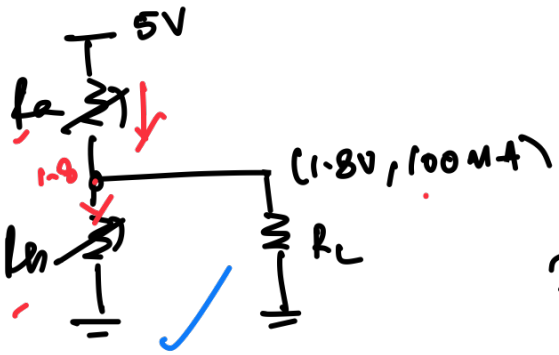
Lecture #22



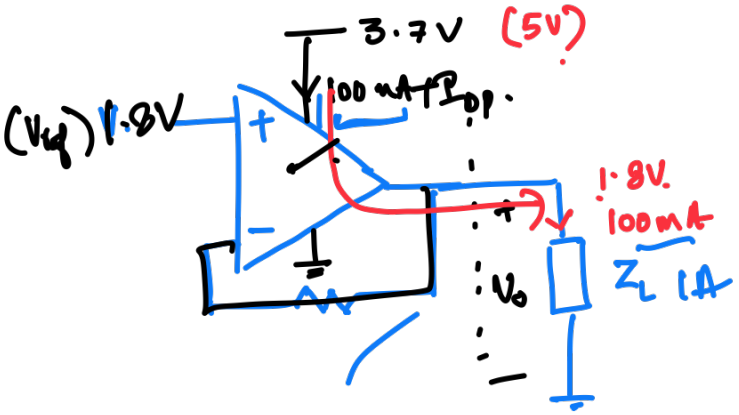
- Phone (3.7V)
  - Display (3.3V)
  - Wireless: 5G - GSM, Bluetooth, NFC, Wifi
  - Wired comm.,
  - Sound
  - Processor. (0.8V, 0.9V)



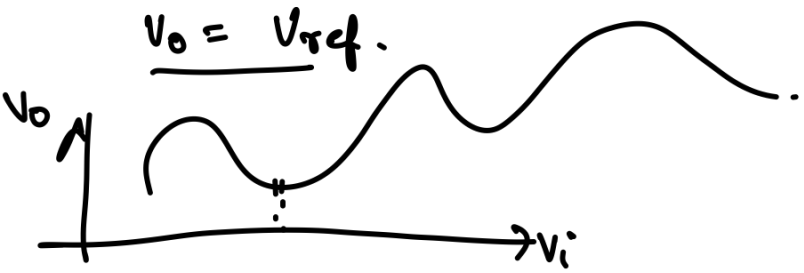
P_L = Power delivered to load = 1.8V × 100mA

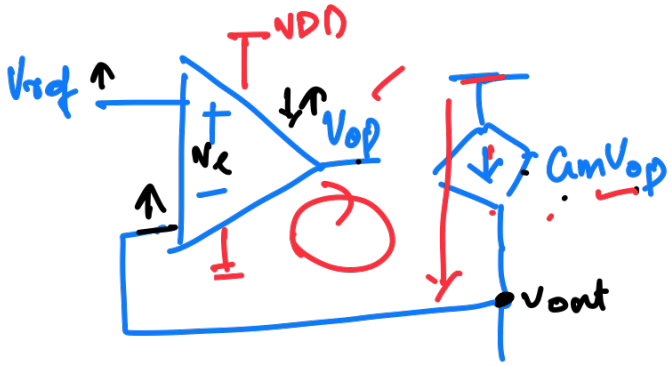
P_S = Power drawn from supply = VDD × (100 + I)

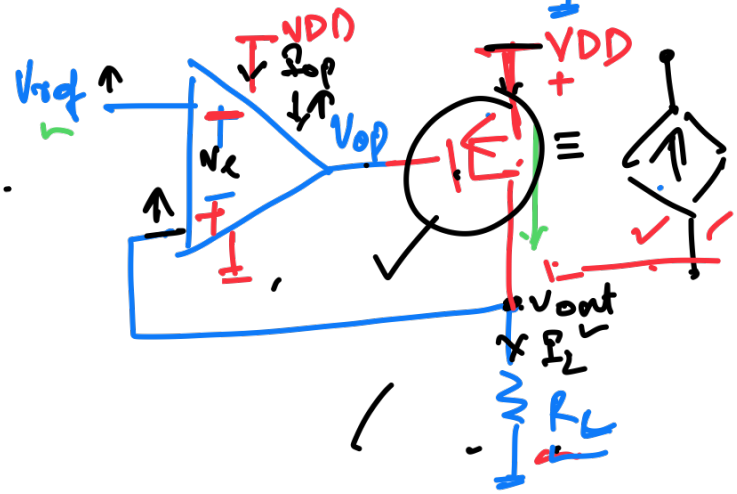
$$\eta = \frac{P_L}{P_S} \times 100\%$$



$\eta < 100\%$

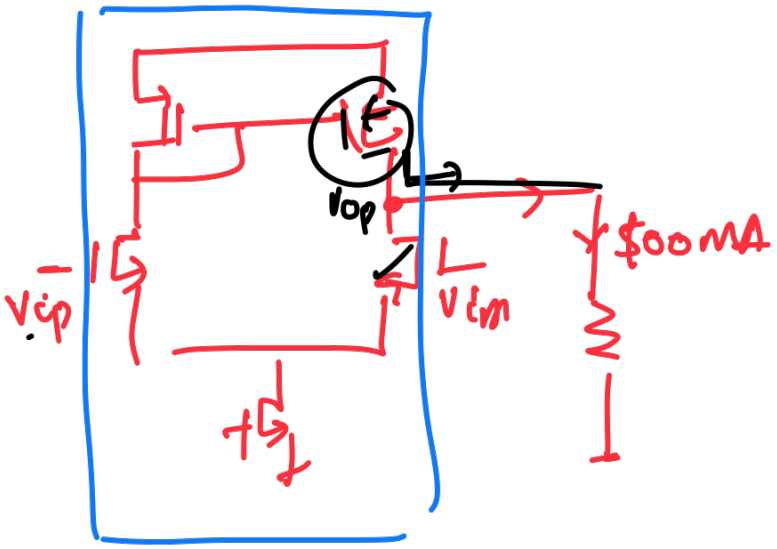


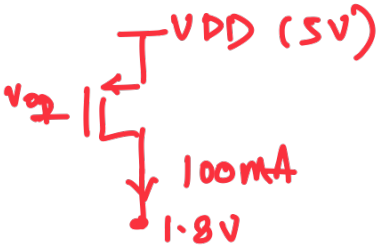




Power delivered to load = $V_{out} \times I_L$ ✓

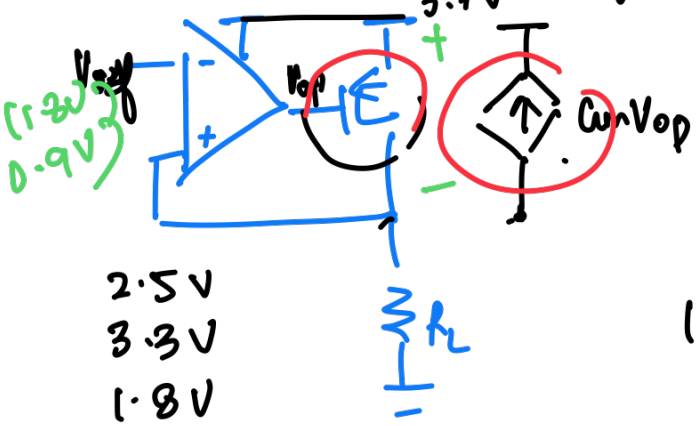
Power drawn from supply = $V_{DD} \times (I_L + I_{op})$

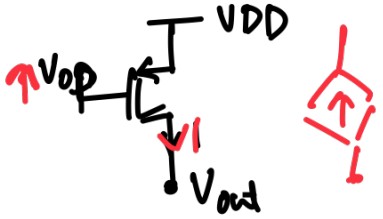






LDO: Low Dropout Regulator.





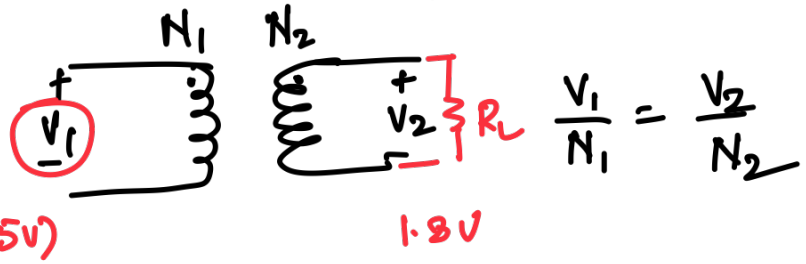
$I = K(V_{DD} - V_{op} - |V_{tp}|)^2$

2.5V
3.3V
1.8V
1.2V
0.9V
0.8V

Power wasted = $(3.7 - 2.5) \times I_{L1} + (3.7 - 3.3) \times I_{L2}$

5.0V ⟶ $1.8 \times I_L = 5.0 \times I_s$

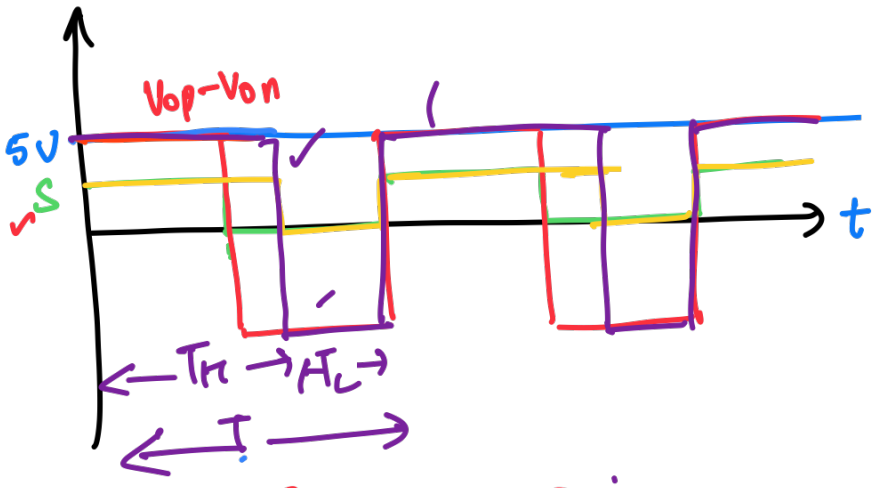
$P_L \quad P_s$



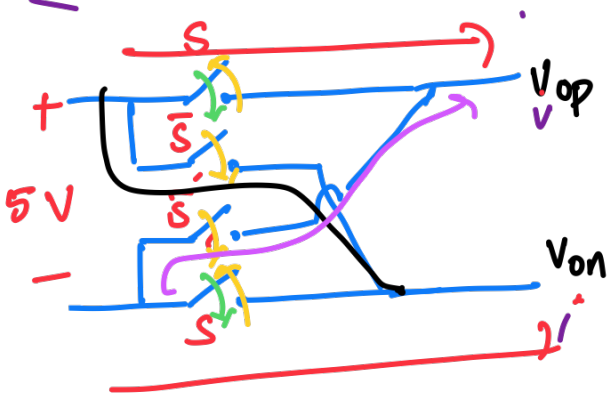
$\frac{V_1}{N_1} = \frac{V_2}{N_2}$

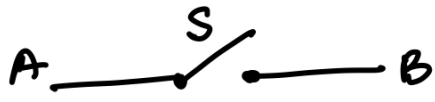
Transformers only with AC.

DC → AC — [Trans.] → AC-DC

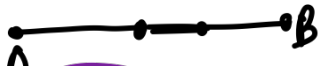


DC o/p voltage = $\frac{T_H}{T} \times 5'$





if S is high:



$R_{SW} = 0$

if S is low:



$R_{SW} = \infty$

